Shashank Rana

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About Me

I am Shashank. Systems, efficiency and scale excite me, in that order.

Currently, I am working at **Google Search** in the RankEmbed, where I am helping RankEmbed scale through growing pains as we strive to serve both the growing AI traffic and an ever-larger share of search traffic.

Academic Details

BITS Pilani Goa Campus (2025)

- B.E. (Hons.) Computer Science
- **CGPA: 9.17**/10.0

Work Experience

Google

Software Engineer | *07/2025 – Present* | *Bengaluru*

- Member of the **RankEmbed** team, owning the system for vector embedding-based retrieval for Search; it is world's largest such system, searching **XXX Bi** documents at **XX Mi** QPS, at **X ms** 95p latency.
- **Automated RankEmbed's eval backend refreshes** : Designed and implemented an agentic pipeline to automate backend refreshes, reducing human toil from **2 SWE days** to **4 hours** and minimizing error.
- **API Migration**: Drove a P0 architectural migration of critical RankEmbed clients (Search, AI Grounding, etc) to a unified API, migrating **X M QPS** worth of clients with **XX M+ QPS** in the pipeline.
  - This involved writing performant code in some of the most complex and hottest critical paths in Google Search, extensively testing and profiling changes to eke out the last ms of latency, while ensuring complete production stability.
  - Transitioned clients from using legacy raw cosine similarity scores to stable percentile conversions offered by the new API, effectively decoupling downstream clients from ML model refreshes.
  - Coordinated with the latency team to minimize migration overhead.
  - **Planned and implemented** a strategy to reduce additional latency of a major migration from **0.4ms to 0.14ms** by breaking the migration into multiple parts.
  - Implemented libraries to enable clients to reconstruct the new API's percentiles to raw scores locally, preventing massive network payload bloat.
- Engineered and deployed a plan to decouple RankEmbed's backend routing from binary rollouts using Google's internal datapush tools.
  - Enabled real-time updates to routing, significantly improving system agility and reducing reaction time for outages from **XX hrs to X mins**.

Harness.io

SWE Intern | *01/2025 – 06/2025* | *Bengaluru*

- Worked on AI Test Automation agent.
- Led migration of long-running tasks from self-managed/self-scaled (EKS) cluster to serverless (AWS Batch).
- Reduced scale-up time from **XX mins to X mins** and enabled infinite scaling.
- Delivered end-to-end features: engineered peripheral services for managing serverless jobs, including a scheduler, sentinel, and validator.

Google

MLE Mentee | *10/2023 – 01/2024*

Participated in a 10-week program focusing on ML Engineering projects.

- Investigated abuse patterns across multiple platforms and aimed to build **resilient cross-platform models** for universal abuse detection. Developed and implemented robust pipelines, **addressing data skew** and managing varied features across platforms.
- Achieved **F1 scores over 0.76** in cross platform tasks, and uncovered how abuse patterns carry across different platforms. Slides.

Courses

Intro to DB systems (**CMU DB** - on YT), Data Storage Technology and Networks, Computer Networks, Operating Systems, Database Systems, Compilers

Research Experience

Data, Systems & High Performance Computing (DaSH) Lab

Undergrad Researcher, BITS Goa (Project FLOps) | *03/2024 – 01/2025*

- Investigated different InfraDrift contributors in FL, in collaboration with Prof. Arnab Paul.
- Proposed strategies to counter them in our publication.
- Designed and managed a heterogeneous FL testbed consisting of 13 nodes.
- Nodes ranged in compute from Nvidia Orin NX to Raspi 400.
- Designed and orchestrated experiments simulating real FL challenges: Data skew, straggler clients, and network heterogeneity.

Technical Proficiency

- **Google Specific:** pprof, Hg/git, anti-gravity SDK (power-user), automon (graphana)
- **Languages:** C++, Python, Java/Kotlin, Picolo, GCL
- **Others:** SQL, DBs, Sockets programming, Raspberry Pi/dev boards, Android development, Git, Angular

Accomplishments and Stats

- **Stats** (Time Period: 11 Months):
  - Delta: **44k**
  - CLs submitted: **133**
  - Bugs closed: **134**
- **Won JPMC Code for Good:** Won the 24-hour JPMC Code for Good Hackathon, which saw participation from over 79 shortlisted teams from across the country.

Teaching

Database Systems

07/2024 – 11/2024